

Notes for 't Hooft Anomaly PTFC Talk.

Sources: "Naturalness, Chiral Symmetry, and Spontaneous chiral symmetry breaking" - 't Hooft (1979)

Weinberg QFT II, Ch 22

Q: Is it possible that some or most of the so called "elementary" fermions (quarks, leptons, etc.) are really bound states of some more fundamental particles?

Immediate trouble: If quarks/leptons are bound up by particles that feel strong but asymptotically free gauge forces, the ^{energy} scale associated with these interactions must be very high ($\Lambda' \gtrsim 10 \text{ TeV}$) to explain our measurements of electric/magnetic moments.

~~Naively~~ Naively we'd expect the bd states of these particles (i.e. the quarks, leptons) to be of the order $m_{bd} \sim 2\pi\Lambda'$ (like in QCD where $m_N \sim 2\pi\Lambda_{QCD}$ where $\Lambda_{QCD} \sim 200 \text{ MeV}$).

But of course most of these particles are significantly lighter! The crux of this ~~problem~~ lies in the fate of chiral symmetry in the 'low energy' theory.

A review of what happens in QCD...

With three very light flavors of quark (u, d, s) (light means $m_q \ll m_N$) the QCD Lagrangian has an approximate chiral ~~flavor~~ symmetry

$$SU(3)_L \times SU(3)_R \times U(1)_V$$

($U(1)_A$ is spoiled by instantons!)

We know this symmetry isn't fully ^{linearly} realized in the IR, because there'd be a parity doubling of all states in the spectrum, which we don't see. We assume due to the strong gauge dynamics that the axial part is spontaneously broken

$$SU(3)_L \times SU(3)_R \times U(1)_V \xrightarrow{XSB} SU(3)_V \times U(1)_V$$

Goldstone: To each of the broken $SU(3)_A$ gens we associate a massless (approx), spinless boson of negative parity (pions, kaons, eta).

* The rest of the b.d states are all quite heavy, getting masses on the order $m \sim 4\pi F_\pi$ or $2\pi\Lambda_{QCD}$ (Here $F_\pi \sim 93 \text{ MeV}$ is the pion decay constant that is effectively the strength of the symm breaking, while $\Lambda_{QCD} \sim 200 \text{ MeV}$ is the scale that emerges when we regularize QCD (dim. transmutation))

XSB is spoiling us having light / massless bound states $\Rightarrow$ what we want is an unbroken chiral symmetry in the IR (naively this prevents mass terms which mix L, R and effects of SSB).

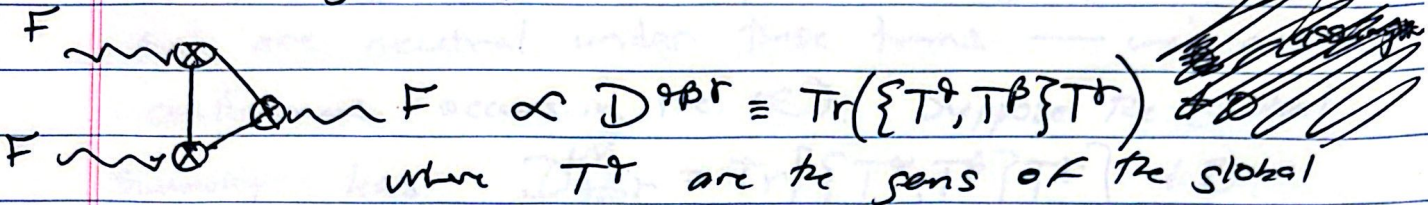
Q: When can / does this actually occur?

t'Hooft: we can use global anomalies to try and answer this question!

— 't Hooft Anomalies + Anomaly Matching

Unlike the previous anomalies we've considered, these 'anomalies' are in triplets of global / flavor currents.

The triangle diagrams look like


$$D^{\alpha\beta\gamma} \propto \text{Tr}(\{T^\alpha, T^\beta\} T^\gamma) \neq 0$$

where T^α are the gens of the global symm on all the L $\bar{H}$ states.

At face value there's nothing wrong with having an 't Hooft anomaly — the currents are conserved unless we couple background gauge fields in.

There is (if $D^{\alpha\beta\gamma} \neq 0$, i.e. "the symmetry has an 't Hooft anomaly") an obstruction to gauging this symmetry, since then we'd have gauge anomalies.

* These anomalies have the remarkable property that they are robust to RG flow — a fancy way of saying that if G_F has 't Hooft anomalies in the UV (and G_F isn't spont. broken) then these anomalies must also exist in the IR.

This is obviously nontrivial if confinement occurs and the IR theory looks nothing like the UV! In that sense it is a powerful tool to make statements about what the IR spectrum looks like.

Why ~~must this happen~~ is this true?

't Hooft's argument: Suppose the LH fermions (and RH antifermions) χ in my UV theory have global chiral symmetries with gens T_a, T_b , etc. (χ we assume also couples to gauge fields that are neutral under these trans — we'll assume confinement occurs in the IR). Suppose the global symmetry has $D_{\text{gpr}}^{\text{UV}} = \text{Tr}[\{T_a, T_b\} T_c] \neq 0$.

• Suppose we wanted to gauge this symmetry. We'd have to add in a set of massless 'spectator' fermions that couple to these new, (presumed) weak gauge bosons. They'd have to have $D_{\text{gpr}} = -D_{\text{gpr}}^{\text{UV}}$ so that all gauge anomalies cancel.

• The IR is defined by the confining phase of the original interactions (the fictitious gauge bosons are still weak here by assumption). The EFT for this system contains the spectator fermions, weak gauge bosons, and any massless b.d. states from our UV fermions and the gluons that bind them. The b.d. states have symm generators $\tilde{T}_a, \tilde{T}_b$, etc. (different from those in the UV), but this EFT must also be anomaly free. The spectators contribute $-D_{\text{gpr}}^{\text{UV}}$, so $D_{\text{gpr}}^{\text{IR}} = \text{Tr}(\{\tilde{T}_a, \tilde{T}_b\} T_c) = -(-D_{\text{gpr}}^{\text{UV}}) = D_{\text{gpr}}^{\text{UV}}$ (the anomalies in the IR match those in the UV).

* This argument goes thru however weak the fictitious gauge ints are, so we don't even need them to conclude that the massless spin $\frac{1}{2}$ b.d. states have the same A-coeffs as in the UV.

Depending on the setup / group in the UV, there are typically many (infinite) solutions for the reps appearing in the IR. — This is sad for our original question regarding composite fermions. As it turns out, the most interesting situation is when the anomaly matching cannot be satisfied $\implies$ there is no solution in terms of reps in the IR that reproduce the UV anom. Then we'd have to conclude that G_F must be spontaneously broken in the IR!

* This is actually the case for QCD with 3 ^{flavors of} massless quark. So, despite the fact we can't seem to use anomalies to figure out what quarks are made of (w/ out extra constraints), we seem to have a 'proof' of XSB in QCD — something as far as I know we simply just assume (there are ad hoc arguments, however). The anomaly matching condition tells us it couldn't be any other way.

Now we have to prove that there are no solutions and actually show how the calculus works.

— The fate of QCD with $N_f = 3$

$$G_C = SU(3)$$

$$G_F = SU(3)_L \times SU(3)_R \times U(1)_V$$

of massless quark flavors.

Baryon #

($U(1)_A$ is spoiled by instantons!)

* Assume first G_F isn't spontaneously broken.

(Technical note: Reps will actually be in $G_F / \mathbb{Z}_3$ since transformations in the center of the color gauge group should be modded out from the flavor group.)

we need at least 3 quarks to form ^{fermionic} color singlet states in the IR.

— UV: q 's are color triplets, and transform in G_F with LH fermions in $(3, 1)$ and RH fermions in $(1, 3)$, both with $U(1)_V$, $\# = 1$.

There are nonvanishing anomaly constants for $(SU(3)_L)^2 U(1)$ (and $SU(3)_R^2 U(1)$) and $SU(3)_L^3$ (and $SU(3)_R^3$)
e.g. $D_L^{ab0} = \frac{1}{3} \delta_{ab}$ — fundamental

Some notation: $\text{Tr} \Pi_R = N(R)$

$$\text{Tr}(T_R^a T_R^b) = T(R) \text{Tr}(T_F^a T_F^b)$$

$$\text{Tr}(\{T_R^a, T_R^b\} T_R^c) = A(R) \text{Tr}(\{T_F^a, T_F^b\} T_F^c)$$

* * When we compute the total anomaly we sum over all LH fermions (helicity $+\frac{1}{2}$), including antiparticles of RH fermion states (helicity $-\frac{1}{2}$).

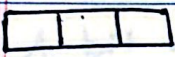
Antifermions transform in complex conj reps, with minus the anomaly constant, so we are effectively counting # of times a rep appears among LH states

— # of times a rep appears among RH states.

(That's also the reason we'll only check the L anomalies)
 To avoid double counting we'll also sum only over
 states with positive baryon #. What reps can appear
 in the IR? Start from

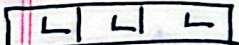
$$3 \otimes 3 \otimes 3 = (6 \oplus \bar{3}) \otimes 3 = (6 \oplus 3) \oplus (\bar{3} \otimes 3) \\ = 10 \oplus 8 \oplus 8 \oplus 1$$

(we can also have combos with mixed chirality)

		N	T	A
3		3	1	1
$\bar{3}$		3	1	-1
10		10	15	27
1		1	0	0
6		6	5	7
8		8	6	0

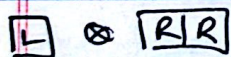
These are
 our building
 block reps.

These are the only reps
 contributing to the L
 anomalies.

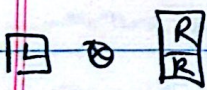
 l_1

 l_2

$L \rightarrow R$
 $\longleftrightarrow$

 l_3

 $-l_3$

 l_4

 $-l_4$

 l_5

* If parity is unbroken then a
 rep (r,s) occurs among $+\frac{1}{2}$ states
 as much as (s,r) occurs among
 $-\frac{1}{2}$ states, hence the minus sign
 for the reps above. *

— $SU(3)_L^3$

$$27l_1 + 0l_2 + (1 \cdot 6 - 7 \cdot 3)l_3 + (1 \cdot 3 + 1 \cdot 3)l_4 + 0l_5 = 3 - N_c.$$

$$\Rightarrow 27l_1 - 15l_3 + 6l_4 = 3$$

— $SU(3)_L^2 U(1)$

$$15l_1 + 0l_2 + (1 \cdot 6 - 5 \cdot 3)l_3 + (1 \cdot 3 - 3 \cdot 1)l_4 + 6l_5 = 1$$

(here we divided by the baryon # contr)

$$\Rightarrow 15l_1 - 9l_3 + 6l_5 = 1$$

A bit of 18th century mathematics ("Bezoo")

Eqs of the form $\sum a_i x_i = q$ where q, a_i are integers have solutions where x_i are integers iff $\gcd(a_1, \dots, a_n)$ also divides q . The set of all possible #s one can form on the RHS are just multiples of the smallest common divisor.

For the second eq, the gcd is 3, so we cannot satisfy this equation! In other words, there is no way to match the UV anomalies with these reps in the IR — actually it's possible to show that this holds for all reps of GF and even when parity is broken. This means chiral $SU(3)$ must be spont. broken!

How do we know its the axial part?

Witten / Vafa: vector symms like $SU(3)_V$ and $U(1)_V$ cannot be spont. broken

(massless ~~the~~^{sb} states made out of massive g 's).

So we must have $SU(3)_L \times SU(3)_R \times U(1)_V$
 $\rightarrow SU(3)_V \times U(1)_V$

Some final points...

- For $N_f = 2$ this argument doesn't actually work — much harder to show that chiral $SU(2)$ alone is spont. broken (we still assume it ^{anyway} in χPT)

- Even when G_F is spont. broken the anomaly matching condition is still satisfied, now by the g_b 's (albeit in a more complicated manner). The g_b s reproduce the anomaly via Wess-Zumino Witten terms ... to be discussed later.